

A
PROJECT REPORT
ON
“MACHINE RENTAL SERVICE”

SUBMITTED TO
SHIVAJI UNIVERSITY, KOLHAPUR
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENT
FOR THE AWARD OF DEGREE
BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND
ENGINEERING

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DKTE SOCIETY'S TEXTILE AND ENGINEERING INSTITUTE,
ICHALKARANJI
(AN EMPOWERED AUTONOUOUS INSTITUTE)
2024-2025

D.K.T.E. SOCIETY'S
TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI
(AN EMPOWERED AUTONOMOUS INSTITUTE)
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CERTIFICATE

This is to certify that, project work entitled

“MACHINE RENTAL SERVICE”

is a bonafide record of project work carried out in this college by

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DECLARATION

We hereby declare that, the project work report entitled “**MACHINE RENTAL SERVICE**” which is being submitted to D.K.T.E. Society’s Textile and Engineering Institute Ichalkaranji, affiliated to Shivaji University, Kolhapur is in partial fulfillment of degree B.Tech.(CSE). It is a bonafide report of the work carried out by us. The material contained in this report has not been submitted to any university or institution for the award of any degree. Further, we declare that we have not violated any of the provisions under Copyright and Piracy / Cyber / IPR Act amended from time to time.

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Thank you,

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ABSTRACT

This project introduces a dynamic and user-friendly Machine Rental Service Website designed to bridge the gap between machine owners and renters. The primary goal is to create a centralized digital platform that simplifies the process of renting and lending both industrial-grade machines and daily-use appliances.

The system allows two types of user roles—Machine Owners, who can register, list their machines with details like type, availability, price, and location; and Renters, who can browse the catalog, search machines by filters, view details, and make bookings. It aims to reduce idle inventory for owners while providing affordable, on-demand access to essential equipment for renters.

Developed using HTML, CSS, JavaScript for the front end and PHP with MySQL for the backend, the platform ensures responsive design, secure login, and easy navigation. Additional features such as a machine exploration dashboard, rental history, and contact management improve user experience.

This system has the potential to benefit industries like construction, agriculture, and home services, and can also be adapted for educational and startup environments. It encourages resource sharing, reduces unnecessary purchases, and promotes

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1.Introduction

In today's fast-paced industrial and construction environments, the need for machinery is indispensable. However, owning heavy or specialized machines is not always a practical or economical option for many businesses, especially small enterprises or contractors who require equipment only for specific durations. The cost of procurement, maintenance, and storage of such machinery poses a major challenge. At the same time, there are numerous machine owners who have idle or underutilized machines that could be rented out to generate additional revenue. This scenario reveals a significant opportunity to establish a service that connects these two parties.

The "Machine Rental Service" project is conceptualized as a web-based platform that bridges the gap between machine owners and potential renters. It allows machine owners to register and list their equipment for rent, providing all necessary details such as machine type, usage, rental terms, and contact information. Renters, on the other hand, can browse the listings, filter according to their needs, and directly contact the machine owner using the information provided on the platform.

A distinctive feature of this system is that it does not involve or manage any financial transactions. The platform purely serves as a directory and communication bridge, thus minimizing legal responsibilities and operational complexities for the service provider. This makes the application lean, efficient, and user-friendly while also encouraging trust and flexibility in negotiation between the two parties.

The platform is built with core functionalities such as user registration, login authentication, machine listing, machine search/filtering, and secure access to contact details. By keeping the application's scope limited to information exchange and excluding payments and logistics, the platform ensures ease of use, data protection, and minimal compliance issues.

This project is especially relevant in a developing economy where maximizing resource utilization and reducing equipment idle time can lead to increased productivity and cost savings. It empowers both small machine owners and renters by providing them with a common digital space to meet their business requirements efficiently.

1.1. Problem Defination:

In various industries such as construction, agriculture, and manufacturing, the need for machines is often temporary and project-based. However, purchasing machinery involves a significant financial investment, which is not always feasible for small businesses or individual contractors. At the same time, many machine owners possess equipment that remains idle for extended periods due to lack of demand or reach. Unfortunately, there is no dedicated and streamlined platform that effectively connects these two parties. Traditional methods like local advertising or personal referrals are inefficient and lack scalability. Existing digital solutions often integrate complex systems like online payments and logistics, which can deter users who prefer direct communication and simple agreements. These platforms also increase liability for the service provider and introduce complications related to transaction handling. Therefore, there is a need for a minimalistic, secure, and user-friendly platform that facilitates machine rentals by enabling direct contact between machine owners and renters. This project addresses that need by creating a web application that excludes payment features and focuses solely on connecting the two parties through contact information exchange.

1.2. Aim and objective of the project:

Aim:

To develop a web-based platform that connects machine owners with renters by providing a space to list and discover machines available for rent, along with their contact details—without handling payments or logistics.

Objectives :

Objectives of project are :

1. To design a user-friendly interface for both machine owners and renters.
2. To allow machine owners to register and post detailed listings of their machines.
3. To enable renters to search and filter machines based on type, location, or availability.
4. To securely store and manage user and machine data.
5. To provide verified contact information to enable direct communication between users.
6. To exclude any payment gateway or transaction handling to maintain simplicity.
7. To ensure data privacy and access control through proper authentication.
8. To minimize legal complexity by limiting the platform's role to information exchange.
9. To support scalability for future enhancements and more machine categories.
10. To build a reliable system that promotes transparency and trust between users.

1.3.Scope and limitation of the project :

Scope of the Machine Rental Service :

1. Machine Listing: Owners can list machines for rent by providing essential details such as machine type, specifications, location, availability, and contact information.
2. Search Functionality: Renters can search for machines based on various criteria, including machine type, location, and availability.
3. Contact Facilitation: The platform provides a way for renters and owners to communicate by sharing contact information for further discussion and negotiation.
4. No Payment Handling: The platform strictly facilitates communication between renters and owners, leaving payment arrangements to be handled outside of the platform.
5. User Authentication: Users (both owners and renters) can register on the platform, create profiles, and manage their listings or rentals.
6. Geographical Coverage: The service can be offered in specific regions or globally, depending on the platform's design and scalability.

Limitations of the Machine Rental Service :

Limitations of the Machine Rental Service are as follows :

1. No Payment Integration: The platform does not facilitate payment processing, making it reliant on external payment methods or trust between renters and owners.
2. Limited Liability: The platform does not assume any responsibility for damages, malfunctions, or disputes between owners and renters.
3. No Rental Agreement: Users must handle their rental agreements privately, as the platform does not provide any legal or formal contract.
4. No Machine Delivery/Collection Service: The platform does not offer services related to the transportation or delivery of machines.
5. Machine Quality Assurance: There is no guarantee or inspection of the machines listed, relying solely on owner-provided information.
6. Limited Customer Support: The platform offers limited support, primarily focusing on user interactions and technical issues rather than handling disputes or complaints.
7. No Feedback System: The platform lacks a feedback or review mechanism for renters and owners, which could help future users make informed decisions.

1.4. Timeline of the project :

The Machine Rental Service project will follow an Agile Software Development approach, with the development broken into several sprints. The project will begin on 25th January 2025 with Sprint 1, focusing on project planning and requirement analysis for two weeks, identifying the system and software requirements, and preparing an initial product backlog. In Sprint 2, from 8th February to 21st February, the team will work on system design and architecture, including database schema and UI/UX mockups. From 22nd February to 21st March, the project will progress through Sprints 3-5, involving iterative development and coding of the

front-end and back-end, with incremental development of core features such as machine listing, search functionalities, and user registration.

Following development, Sprint 6 and 7 will focus on testing from 22nd March to 4th April, including unit testing of individual modules, integration testing, and system validation to ensure the platform meets the specified requirements. In Sprint 8, from 5th to 11th April, the team will prepare for the system's deployment and launch, conducting final checks and preparing documentation. The project will conclude with Sprint 9, from 12th April to 25th April, focusing on post-launch support and maintenance, monitoring the system for any issues, gathering user feedback, and addressing any necessary fixes or improvements. This Agile approach ensures continuous development and adjustment based on feedback, delivering incremental, functional parts of the system for early testing and evaluation.

1.5. Project Management Plan :

Task	Duration	Start Date	End Date	Assigned To	Priority
Requirement Gathering	7 Days	25 Jan. 2025	31 Jan. 2025	All team Members	High
Product Backlog Finalization	7 Days	1 Feb. 2025	7 Feb.2025	All team Members	High
UI/UX Design	7 Days	8 Feb. 2025	14 Feb. 2025	All team Members	Medium
Data base And Architecture Setup	7 Days	15 Feb. 2025	21 Feb. 2025	All team Members	High
Core Module Development	14 Days	22 Feb. 2025	7 Mar. 2025	All team Members	High
Additional Features	14 Days	8 Mar. 2025	21 Mar. 2025	All team Members	Medium
Unit and Integration Testing	7 Days	22 Mar. 2025	28 Mar. 2025	All team Members	High
System Testing	7 Days	29 Mar. 2025	4 April 2025	All team Members	High
Deployment Preparation	7 Days	5 April 2025	11 April 2025	All team Members	High
Final Testing and Bug Fixing	7 Days	12 April 2025	18 April 2025	All team Members	Medium

Post Launch Support	7 Days	19 April 2025	25 April 2025	All Team Members	Medium
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1.6. Project Cost :

The software is built using open-source technologies like HTML, CSS, JavaScript, PHP, and MySQL, and no additional hardware is used other than the existing computer systems. Hence, the cost of the project mainly includes the machine (laptop/PC) and basic utility expenses.

The project cost is estimated using the COCOMO (Constructive Cost Model) approach.

COCOMO Model Calculations:

Effort Calculation (E):

$$E = 10 \times (0.950)^{1.5}$$

$$E \approx 10 \times 0.922$$

$$E \approx 9.22 \text{ person-months}$$

Development Time (D):

$$D = 2.5 \times (9.22)^{0.9}$$

$$D \approx 2.5 \times 8.28$$

$$D \approx 20.7 \text{ months}$$

People Required (P):

$$P = (9.22 / 20.7) \times 10$$

$$P \approx 4.45 \text{ persons}$$

Assumptions:

- Constant: Semi-detached mode value of 3 (based on 1998 COCOMO standards)
- Estimated Size: 950 LOC (Lines of Code)
- Scale Factor: Moderate (based on project complexity)
- Effort Multiplier: Average values assumed
- Total Functional Points: 6 points
- Conversion Rate: Functional points to LOC considered based on typical standards

Hardware/Software Cost (Approximate in INR):

Component | Estimated Cost (₹)

| Computer System (i5/i7, 8GB RAM or higher) | 20,000 |

| Web Development Environment (VS Code, XAMPP) | 0 |

| 8GB-16GB RAM upgrade (if needed) | 7,000 |

| Electricity Charges (for development time) | 1,000 |

| Internet Charges (for online hosting, testing) | 1,200 |

| Domain and Hosting (optional for deployment) | 2,000 (basic plan) |

Summary:

Total Software Development Cost: Primarily developer time (human resource)

Total Hardware/Utility Cost: Approx. ₹30,000 to ₹32,000 depending on setup

License Costs: ₹0 (as open-source software is used)

Deployment Cost: Optional (₹2,000 for domain + hosting)

Thus, the project is highly cost-efficient for college or startup level deployment.

2. Background Study And Literature Overview

2.1. Literature Overview:

The demand for flexible access to machinery and equipment has significantly increased in recent years due to the rapid growth of industries, startups, and evolving lifestyles. Traditional ownership models are being replaced with more sustainable and cost-effective solutions such as rental services. This shift has led to the emergence of online platforms that connect machine owners with renters, facilitating the temporary use of industrial and household machines without the burden of full ownership.

Several platforms globally, such as YoRent and EZRentOut, have demonstrated the practicality and commercial success of rental ecosystems. These systems typically provide features like product listings, user registration (as owner or renter), booking, payment gateways, and reviews. However, many of these platforms are tailored to niche markets or specific equipment categories.

This project aims to create a comprehensive and user-friendly web application that supports both industrial machinery (e.g., drilling machines, forklifts) and daily-use appliances (e.g., vacuum cleaners, water purifiers). By allowing users to register either as machine owners or renters, the system facilitates a decentralized model for equipment sharing. Owners can upload machine details, while renters can explore available options and book as per their needs.

The backend architecture ensures secure data handling, including user authentication, machine listing, and dynamic role-based redirection. Frontend enhancements such as modals, interactive buttons, and clean layouts make the system intuitive and accessible.

This system contributes to the circular economy by promoting reuse, reducing idle resources, and enabling cost-effective access to machinery for individuals and businesses alike.

2.2. Critical appraisal of other people's work:

Several platforms have attempted to digitize and streamline the machine rental process. The most prominent among them include YoRent, EZRentOut, and RentSher. While these platforms are functional and have captured market segments effectively, a closer examination reveals some limitations and areas of improvement.

Equipment Rentals India (ERI)

A platform facilitating the sale, purchase, and rental of construction equipment across India.

Features:

Offers earthmoving, road construction, concreting, and lifting equipment.

Allows contractors to list equipment for rent or sale.

Provides a marketplace for buyers and renters to connect.

Target Users: Contractors and construction companies.

Reference: equipmentrentalsindia.com

Parishi Rental Services

Specializes in construction equipment rental services across India.

Features:

Maintains a fleet of over 1,000 machines.
Focuses on providing high-quality equipment for construction projects.
Target Users: Construction industry professionals.
Reference: parishirental.com

TransRentals

Offers a wide range of equipment rental services, including construction and logistics equipment.
Features:
Serves over 150 locations in India.
Collaborates with more than 5,000 vendors.
Provides quotes and connects renters with equipment providers.
Target Users: Construction companies and logistics providers.
Reference: transrentals.in

JFarm Services

A platform connecting farmers with farm machinery owners for equipment rental.
Features:
Operates in 16 Indian states.
Has impacted over 3 million farmers since 2017.
Facilitates direct negotiation between farmers and equipment owners.
Target Users: Farmers and agricultural equipment owners.
Reference: jfarmsservices.in

Rentkart

A B2B equipment rental and leasing aggregator in India.
Features:
Provides access to thousands of equipment items.
Connects businesses with premier suppliers.
Target Users: Businesses requiring equipment for various projects.
Reference: rentkart.com

FARMS (Farm Machinery Solutions) – Government of India

Developed by the Ministry of Agriculture & Farmers Welfare, FARMS is a multilingual mobile application designed to facilitate the hiring of agricultural machinery across India.
Key Features:
Connects farmers with Custom Hiring Centers (CHCs), Farm Machinery Banks (FMBs), and Hi-tech Hubs.
Supports 12 regional languages for broader accessibility.
Enables listing and renting of various agricultural equipment.
Provides a platform for buying and selling used farm machinery.

Reference: FARMS App Official Website

While existing solutions are rich in features, they often fall short in affordability, ease of use,

and inclusivity for smaller renters or individual owners. The proposed system aims to address these gaps by offering a lightweight, accessible, and flexible solution tailored to a broader audience with a more localized focus.

2.3. Investigation of Current Project and Related Work :

Introduction to the Current Project

The project under investigation is a Machine Rental Service Platform designed to bridge the gap between machine owners and short-term renters. The system operates without a built-in payment gateway, instead focusing on transparency, ease of communication, and categorization of machines across diverse sectors including agriculture, construction, logistics, and domestic appliances.

The objective of the platform is to:

Simplify the process of renting machinery for temporary use.

Allow machine owners to generate revenue from underutilized equipment.

Enable renters to access machinery without the burden of full ownership.

Provide a clean, contact-enabled interface with direct communication, scheduling features, and trust mechanisms like verification and reviews.

Related Work and Existing Solutions

Several platforms exist that aim to address similar problems, particularly within the agricultural and construction domains. A detailed review of these platforms is provided below:

a. FARMS App (Government of India)

Scope: Agriculture only.

Strengths: Government backing, multilingual support, integration with CHCs.

Limitations: No direct contact between parties, no real-time availability, limited to farm machinery.

Comparison: Unlike FARMS, the current project covers multiple industries and facilitates direct renter-owner interaction.

b. JFarm Services (TAFE)

Scope: Agriculture-based P2P rental system.

Strengths: Simplicity, farmer-focused services.

Limitations: Lack of availability calendar, absence of cross-sector machinery.

Comparison: The current project extends beyond agriculture and introduces structured scheduling and feedback systems.

c. KhetiGaadi

Scope: Tractor and farm equipment marketplace.

Strengths: Regional language support, large equipment inventory.

Limitations: Overwhelming user interface, no centralized rental workflow.

Comparison: The current platform emphasizes UI simplicity and streamlines the rental process with availability tracking.

d. Rentkart, TransRentals, and other construction platforms

Scope: Focused on logistics and construction.

Strengths: Category-specific inventory and availability.

Limitations: Limited to commercial use; lacks flexibility for small/individual renters.

Comparison: The proposed platform is designed for both commercial and personal rentals, with a broader machine category base.

Current Project Overview:

The current project aims to develop an online platform where users can rent or list industrial and daily-use machines. The system distinguishes between two user roles: Renter (person who wants to rent machines) and Owner (person who owns and lists machines). Key features include user authentication, machine listing, machine browsing, booking management, and admin control.

How the Current Project Is Different:

Tailored specifically for both industrial and daily-use machines.

Clean and simple UI/UX for students, small businesses, and local vendors.

Lightweight and easy to deploy on college or startup scale servers.

Focused on educational learning, demonstrating full-stack development and database integration.

3.Requirement analysis

3.1. Requirement Gathering:

The Machine Rental Service is a web-based platform designed to bridge the gap between machine owners and renters. The system allows machine owners to list their machines for rent, while renters can browse and contact the owners directly. This platform strictly focuses on listing and communication, and does not handle any payment processing.

1. Functional Requirements

These are the core features the system must offer:

User Registration and Login:

Owners and renters can create accounts and log in securely.

Machine Listing (Owner Module):

Owners can upload machine details including:

Machine name

Description

Category/type

Rent price (per day/week/month)

Location

Contact information (e.g., phone number, email)

Images of the machine

Machine Browsing and Search (Renter Module):

Renters can:

View available machines

Search machines by location, category, or rent range

View machine details and owner's contact info

Contact Interface:

The platform will display the machine owner's contact details.

Renters initiate direct communication (via call/email) outside the platform.

Admin Panel (Optional):

Admin can manage listings, users, and monitor platform usage.

2. Non-Functional Requirements

Usability:

Clean and user-friendly interface

Responsive design for mobile and desktop

Security:

Passwords should be stored securely (e.g., hashed)

Basic input validation and sanitization to prevent attacks

Performance:

Fast search and browsing of listings

Scalable backend for handling more users over time

Scalability:

Capable of expanding to include additional features like reviews or ratings in the future

3. Stakeholders

Machine Owners: Individuals or businesses who want to rent out their machines.

Renters: Individuals or businesses who want to rent a machine for a specific purpose.

Admin (Optional): Maintains the platform's listings and user content.

4. Tools and Technologies

Frontend: HTML, CSS, JavaScript (or frameworks like React if applicable)

Backend: C++ (for logic if required), or Java/PHP/Python

Database: MySQL or SQLite for storing users and listings

Hosting: Localhost for demo or web hosting service for deployment

3.2. Requirement Specification:

No	Requirement	Essential/Desirable	Description of Requirement
RS1	The system should allow users to register and log in	Essential	Users should be able to create accounts as owners or renters and log in securely
RS2	The system should allow owners to list machines for rent	Essential	Owners should be able to add machine details like name, description, rent, location, etc.
RS3	The system should allow renters to browse and search machine listings	Essential	Renters should be able to view all machines, filter by location/category, and search easily
RS4	The system should display machine owner contact details to renters	Essential	Upon selecting a machine, renters should see contact info to communicate directly
RS5	The system should	Essential	Owners should

	allow owners to update or delete their listings		manage their posted listings through their dashboard
RS6	The system should provide an intuitive and responsive user interface	Desirable	The interface should be user-friendly and function smoothly across devices
RS7	The system should securely store user credentials	Essential	Passwords must be hashed and all user data protected with basic security practices
RS8	The system should validate all input fields to prevent invalid entries	Essential	Input validation must be applied for forms like registration and machine posting
RS9	The system should be modular and scalable for future enhancements	Desirable	The system design should allow easy addition of features like reviews or payment options

3.3. User Story :

Renter:

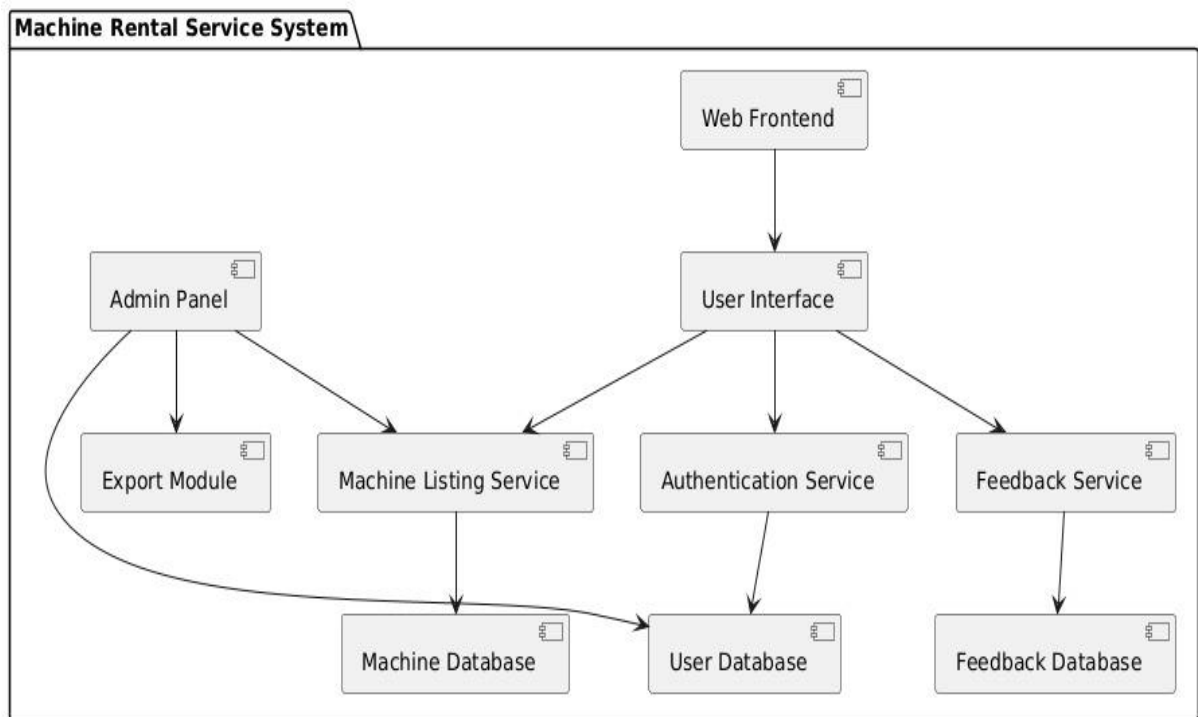
1. User should be able to log in.
2. User should be able to update his profile.
3. Now user has to choose the category of machine that he wants.
4. User should be able to search machine.
5. List of searched machines in that category will be displayed with machine availability, owner's details, location.
6. User has to select a particular owner, so user gets details about the owner (name, contact number).
7. If the selected option is convenient to the user, then he will make contact with that owner.
8. At the end, user should be able to give reviews to the machine owner.
9. User can cancel the machine.

Machine Owner :

1. Machine owner should be able to log in.
2. Machine owner should be able to update their profile (name, contact details, business details).
3. Owner should be able to add machine details (category, specification, rental price & availability).
4. Owner should be able to update or delete machine details.
5. Owner should be able to mark a machine as available/unavailable.
6. Machine owner should receive requests from users who want to rent a machine.
7. Owner should be able to view user's details.
8. If owner agrees, they can proceed with the booking & communicate with user.
9. Machine owner should be able to receive reviews from users after rental.
10. The machine owner should be able to cancel a booking if necessary and notify the user

4. System Design

4.1. Architectural Design :



1. Web Frontend

Provides a user interface for owners, renters, and admins.

Allows role-based access: post machines, search listings, admin monitoring.

2. User Interface

Interacts with backend services through HTTP requests.

Handles form submissions, user login, and feedback input.

3. Authentication Service

Verifies user credentials and manages sessions.

Role-based redirection (Owner, Renter, Admin).

4. Machine Listing Service

Allows owners to post, update, and delete machine listings.

Enables renters to view, search, and filter machine listings.

5. Feedback Service

Collects and stores renter feedback about machine listings or services.

Links feedback to specific machines and renters.

6. Admin Panel

Used by admin to:

View/manage user details and listings.

Monitor system activity.

Export system data.

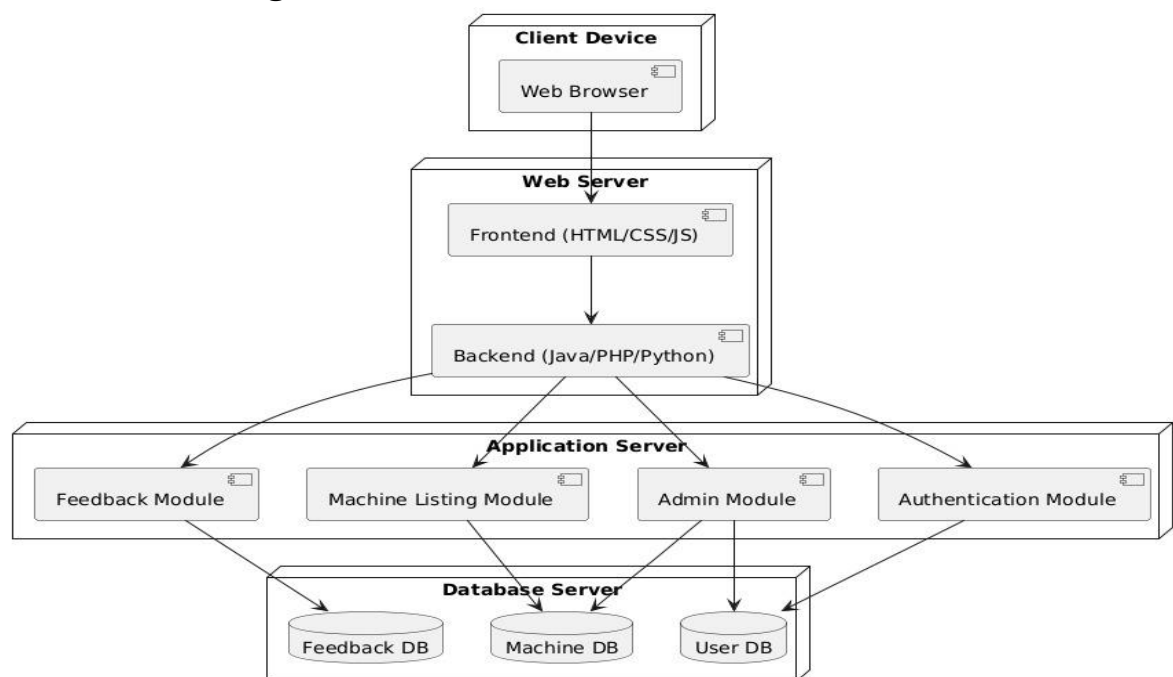
7. Databases

User Database: Stores all user-related data (credentials, roles).

Machine Database: Stores machine listings and owner IDs.

Feedback Database: Stores feedback and ratings by renters

4.2. Software Design Model :



4.3. Algorithmic Description Of Each Model :

1. Authentication Module

Purpose: Manages user registration, login, and role selection.

Algorithm:

1. Display login or registration option.
2. If registration selected:
Take user details (name, email, password, role).
Check if email already exists in User DB.
If not, encrypt password and store user data.
Display success message.
3. If login selected:

Take email and password.
Verify credentials with User DB.
If valid, retrieve role and redirect to corresponding dashboard.
Else, show error.

2. Machine Listing Module

Purpose: Allows machine owners to post, edit, and remove machine listings.

Algorithm:

1. Owner logs in and accesses dashboard.
2. Select "Add Machine":
Enter machine name, type, price, location, contact info.
Generate unique machine ID.
Store details in Machine DB linked with owner ID.
3. To edit/remove:
Owner selects machine from list.
Perform edit/delete operation.
Update DB accordingly.
4. Renter-side:
Search/filter machines using type/location.
Display results dynamically from Machine DB.

3. Request Machine / Contact Info Module

Purpose: Renter sends a request to rent a machine and gets the owner's contact.

Algorithm:

1. Renter browses machine list.
2. Select a machine to request.
3. System logs request with machine ID and renter ID in Request DB (optional).
4. Retrieve owner's contact info from Machine DB.
5. Display contact info to renter.

4. Feedback Module

Purpose: Collects renter feedback on machines or experience.

Algorithm:

1. Renter logs in and selects a previously requested machine.
2. Clicks "Give Feedback".

3. Inputs rating and text feedback.

4. System stores

Feedback ID (auto-generated)

Machine ID

Renter ID

Feedback content

5. Machine Owner views feedback from dashboard.

5. Admin Module

Purpose: Allows admin to view system activity and export data.

Algorithm:

1. Admin logs in using secure credentials.

2. Select dashboard options:

View all users → Fetch from User DB.

View all machines → Fetch from Machine DB.

View feedback → Fetch from Feedback DB.

3. To export data:

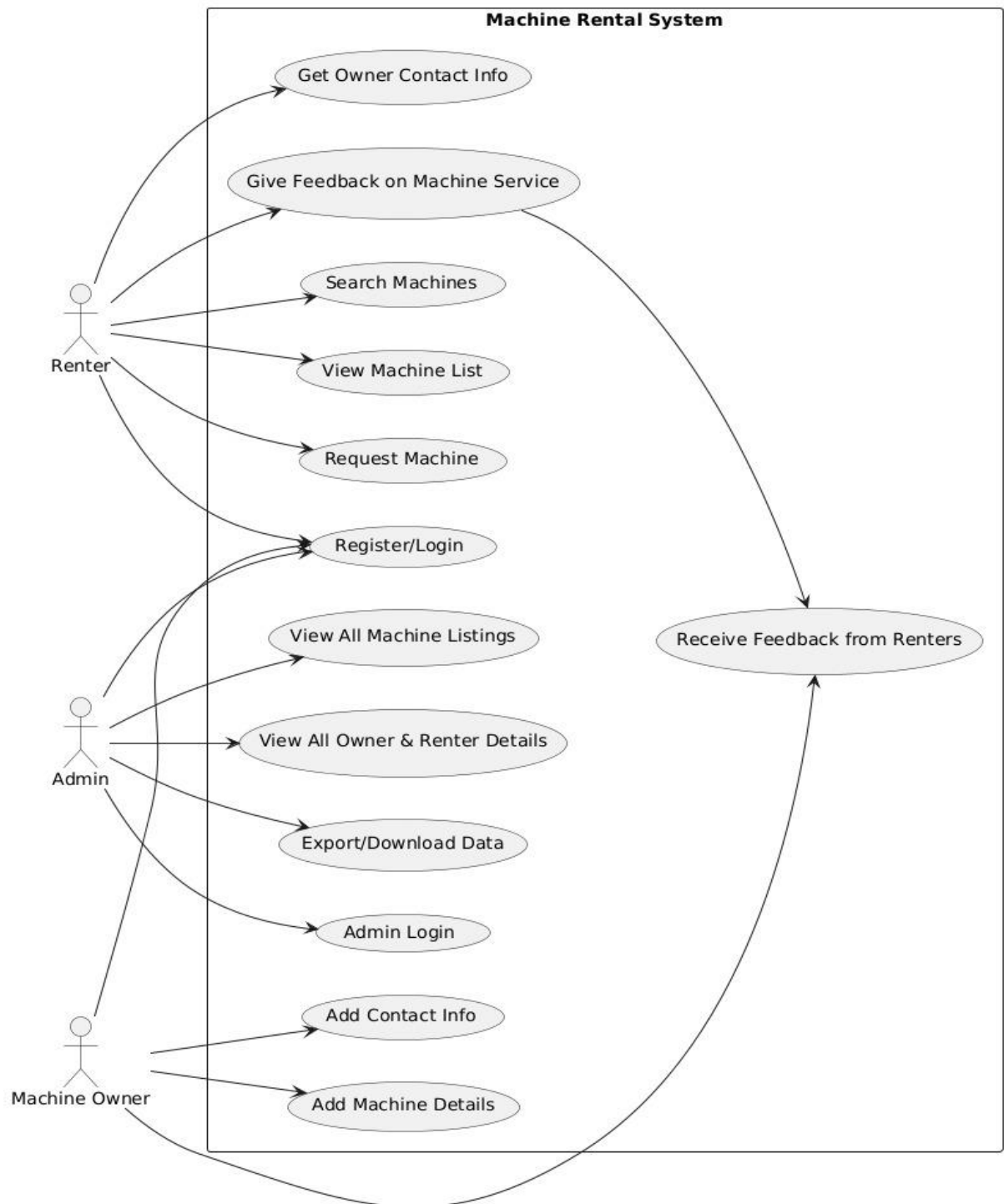
Select export type (CSV, PDF).

Generate report using selected tables.

Offer download option.

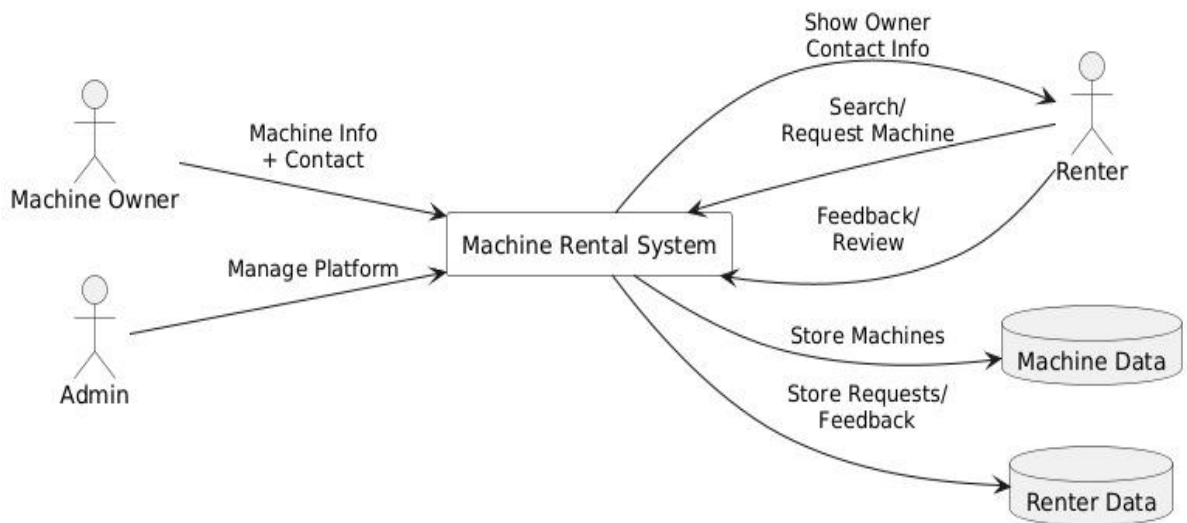
4.4. System Modeling :

4.4.1 Use Case Diagram :

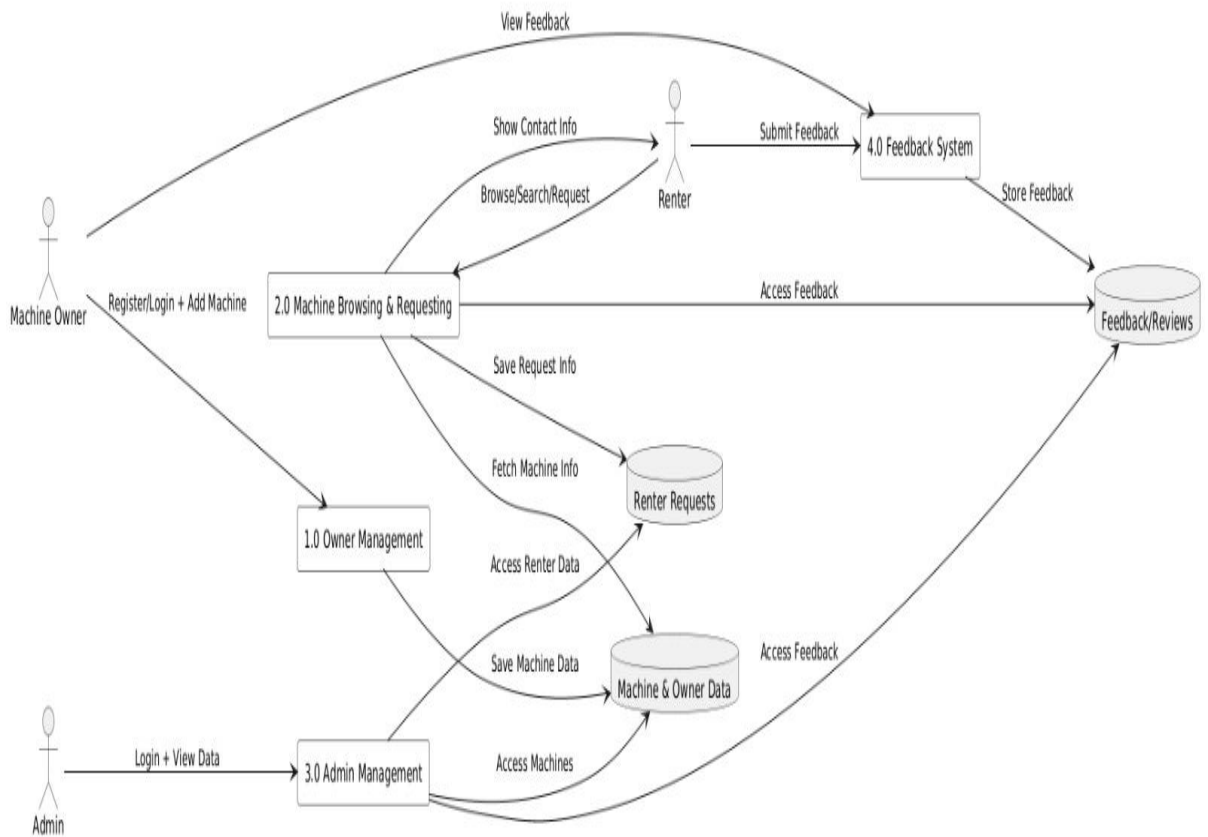


4.4.2 Data Flow Diagram :

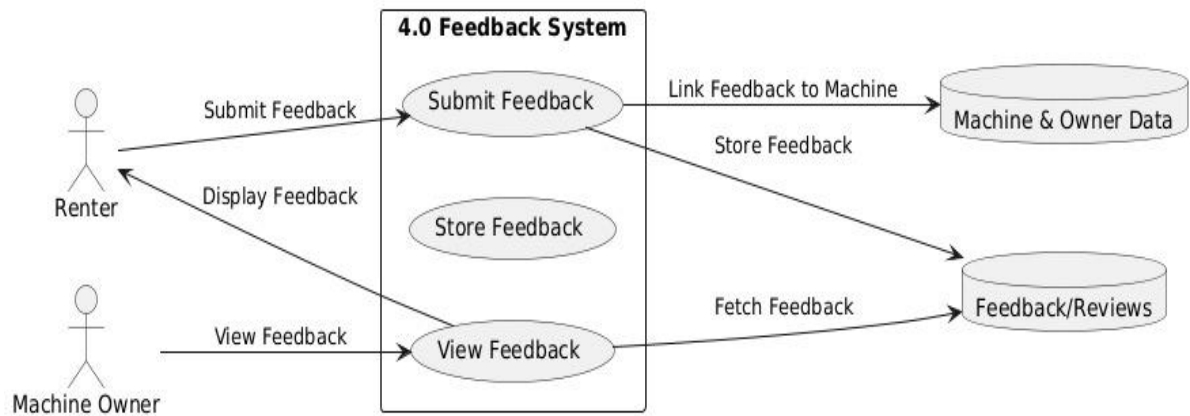
Level 0 :



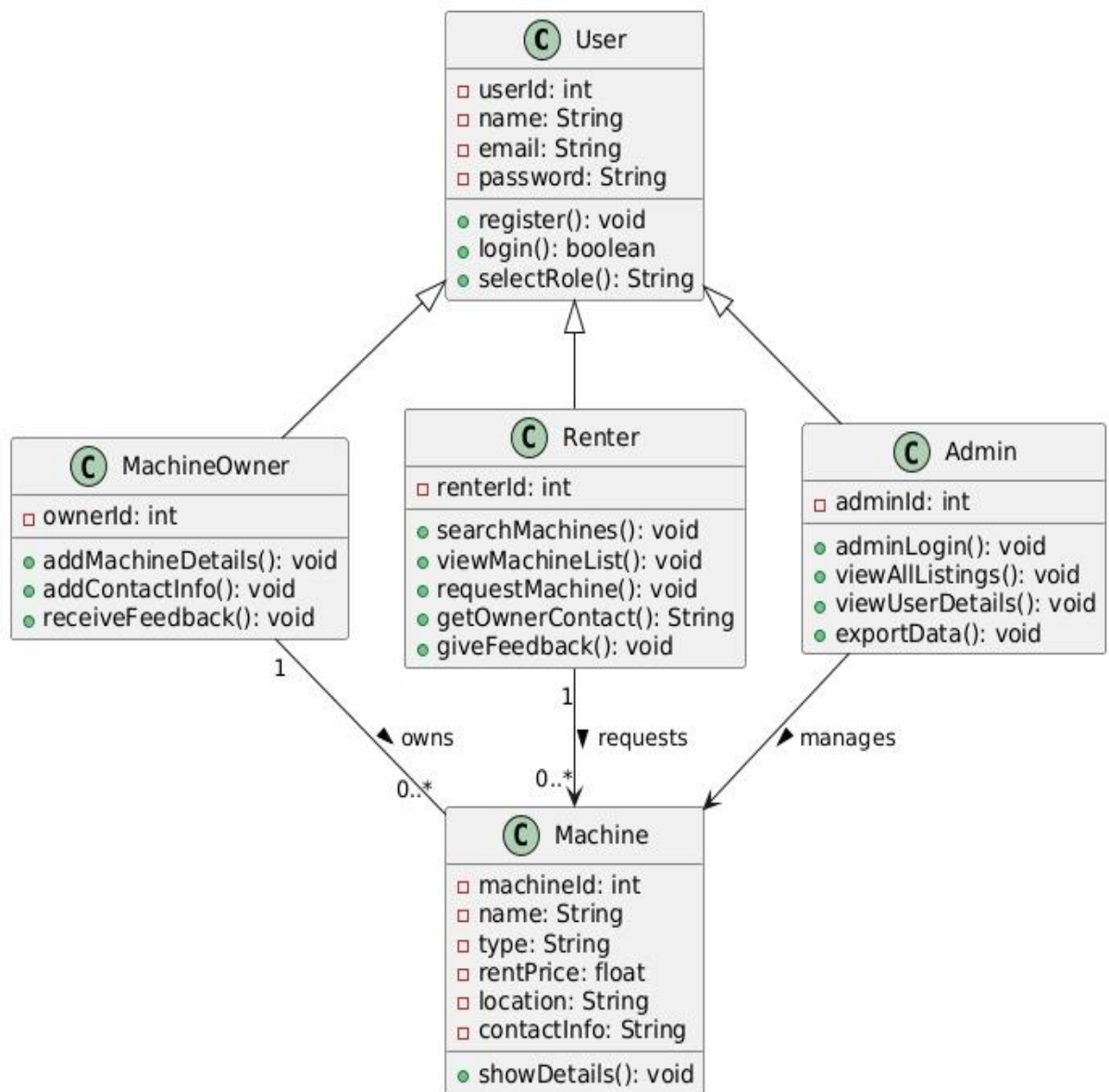
Level 1:



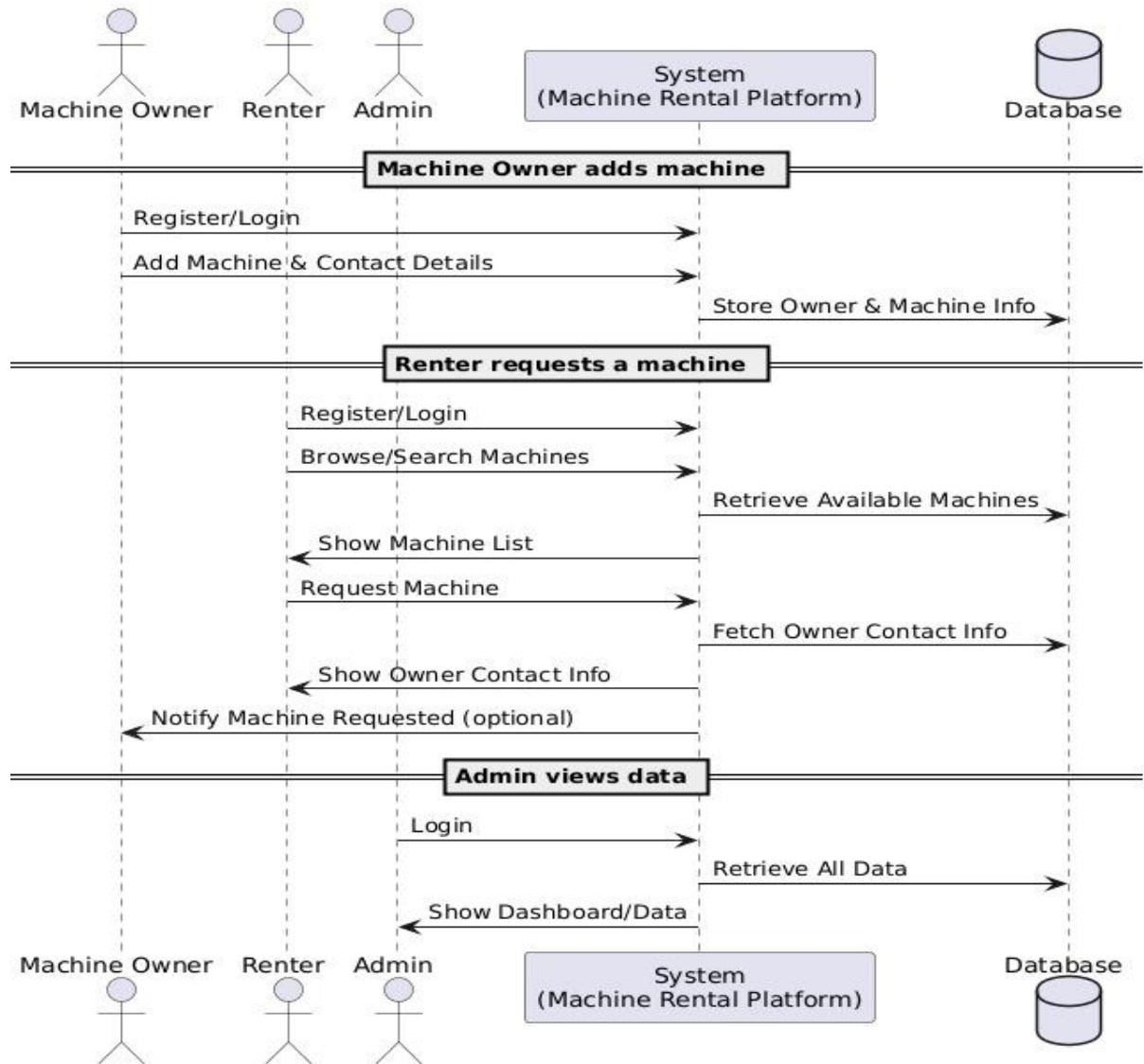
Level 2 :



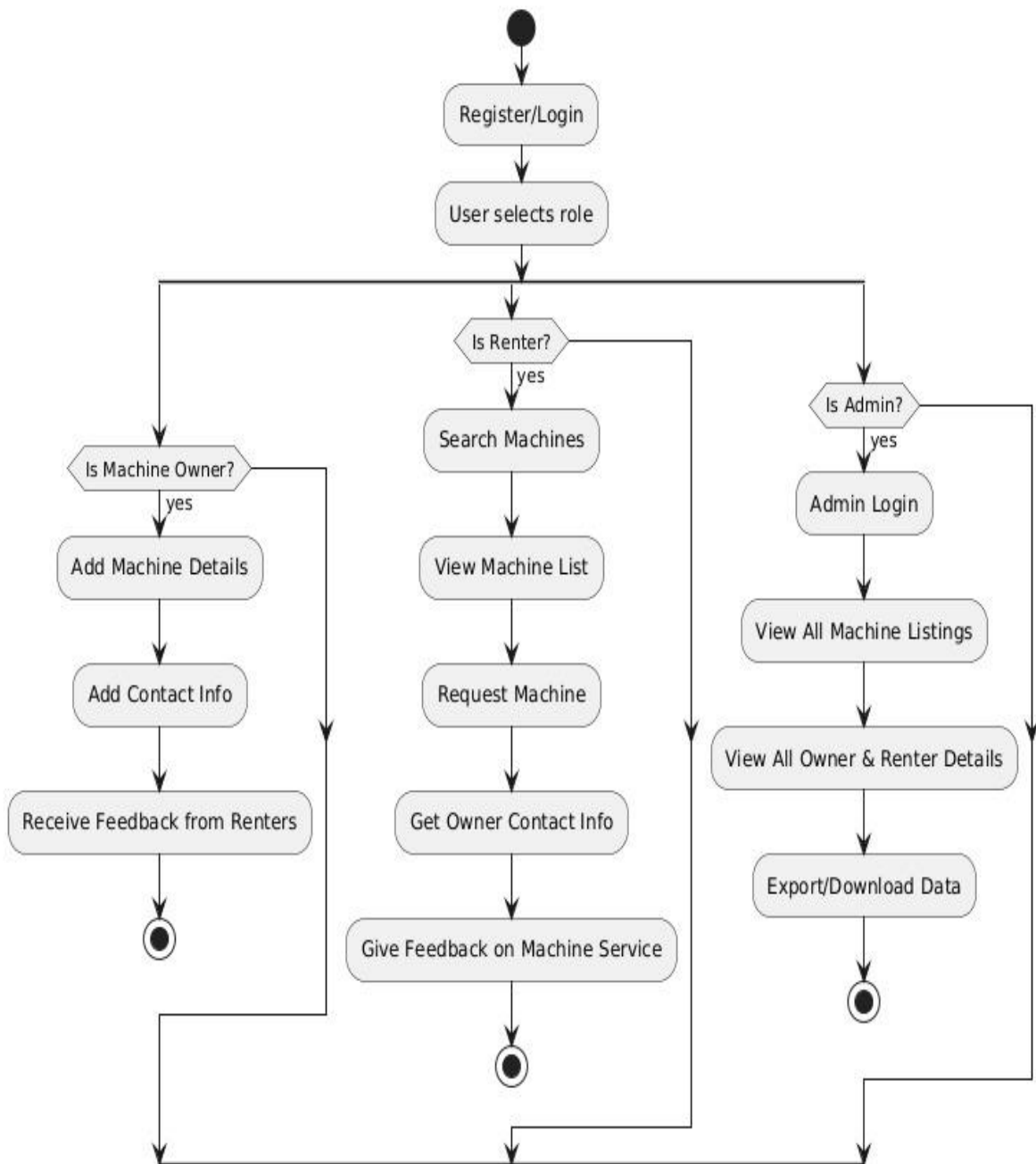
4.4.3 Class Diagram :



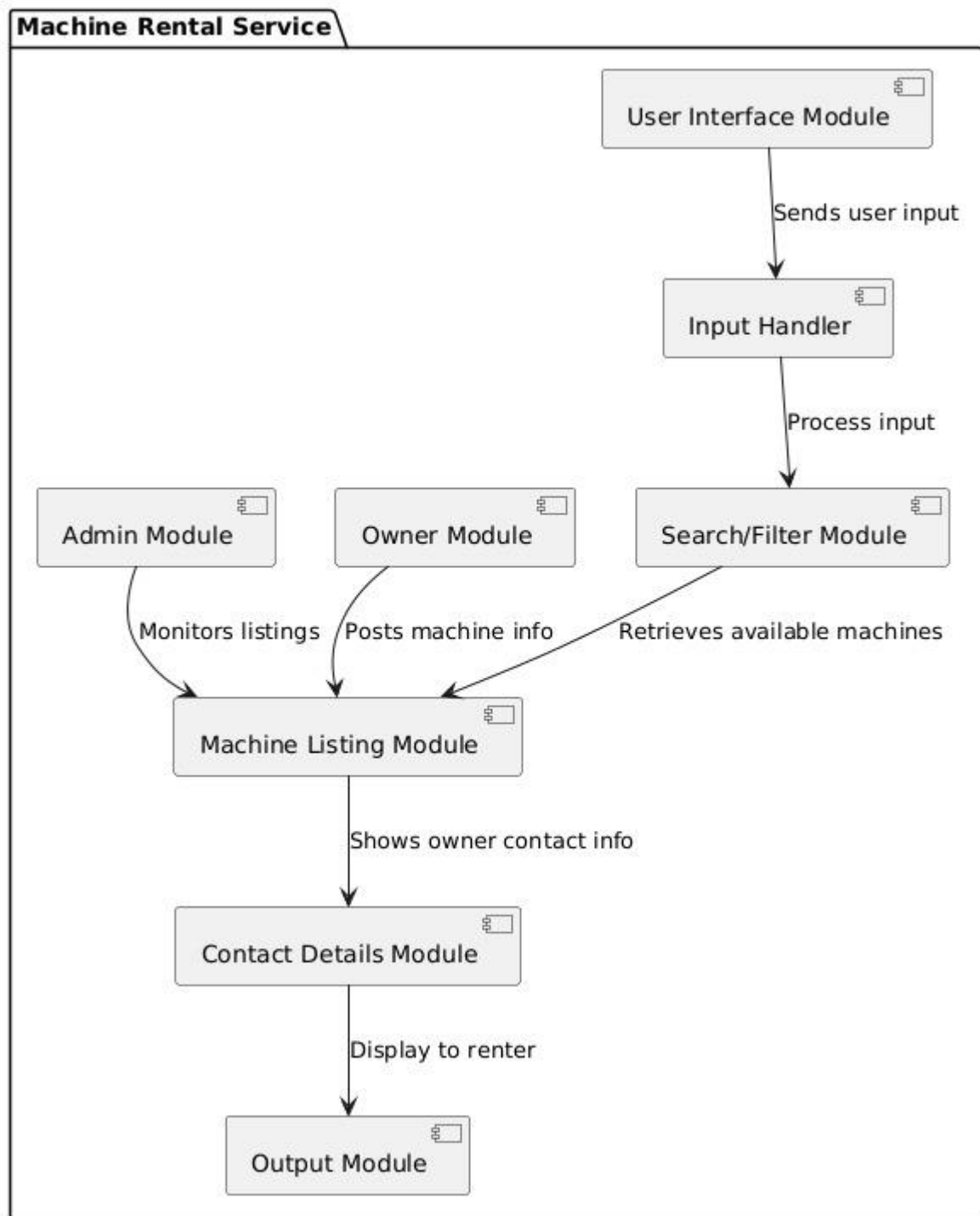
4.4.4 Sequence Diagram :



4.4.5 Activity Diagram :



4.4.6 Component Diagram :



5.Implementation

5.1. Environmental setting for Running the Project:

The environmental setting refers to the infrastructure, tools, and technologies required to develop, deploy, and run the Machine Rental Web Application efficiently and securely. Below are the key components:

1. Hardware Environment

Server Machine (for deployment): Minimum 4GB RAM, 2-core processor (e.g., DigitalOcean, AWS EC2).

Client Devices: Compatible with desktops, tablets, and smartphones with internet access.

Development Machines: Laptops or PCs with minimum 8GB RAM for developers.

2. Software Environment

Frontend:

HTML5, CSS3 (internal or external), JavaScript
Optional Frameworks: Bootstrap (for responsiveness)

Backend (recommended options):

Option 1: PHP with MySQL
Option 2: Node.js with MongoDB
Option 3: Python (Flask/Django) with PostgreSQL

Database:

Relational: MySQL/PostgreSQL (for structured user and machine data)
NoSQL: MongoDB (optional, for flexible machine listing and rentals)

Web Server:

Apache or NGINX
IDE/Code Editors:
VS Code, Sublime Text, or JetBrains IDEs

3. Hosting & Deployment

Local Testing: XAMPP, WAMP, or Local Server setup
Live Hosting: GitHub Pages (static), Firebase (basic), or Heroku/Vercel (dynamic)

Domain Name: Purchased via GoDaddy or Namecheap (optional)

4. Security & Performance

HTTPS with SSL certificates (Let's Encrypt)
Basic input validation and form sanitization
Login authentication (e.g., session-based or JWT)

5. Network Requirements

Stable internet connection for access to hosted platform
Firewall configuration on server for secure access

6. Stakeholder Usage Environment

Owners: Should be able to list machines with images and specifications.

Renters: Should be able to browse, filter, and book machines based on location and availability.

5.2. Detailed Description of Methods :

Project Setup & Functional Workflow – Step-by-Step

Step 1: Visit the Website

User opens the home page.
Sees options like Login and Sign Up.

Step 2: User Sign Up

Click on Sign Up.
Chooses role:
Machine Owner — can add machines.
Renter — can browse and rent machines.
Fills in details: name, email, contact number, city, password.
Data is stored in the database.

Step 3: User Login

User logs in with email and password.
System checks login credentials from the database.
If correct:
Owner is redirected to Owner Dashboard.
Renter is redirected to Machine Explore Page.

Step 4: Owner Dashboard

Owner sees:

Option to Add Machine (name, type, photo, location).

View and manage list of machines uploaded.

Step 5: Renter Dashboard

Renter sees:

Explore Machines button.

Machine listings with filters like type, city, and availability.

Option to Book a Machine.

Booking request is sent/stored in the system.

Step 6: Booking Confirmation

Booking info is saved in the bookings table in database.

Optionally:

Renter sees “My Bookings.”

Owner sees “Requested Rentals.”

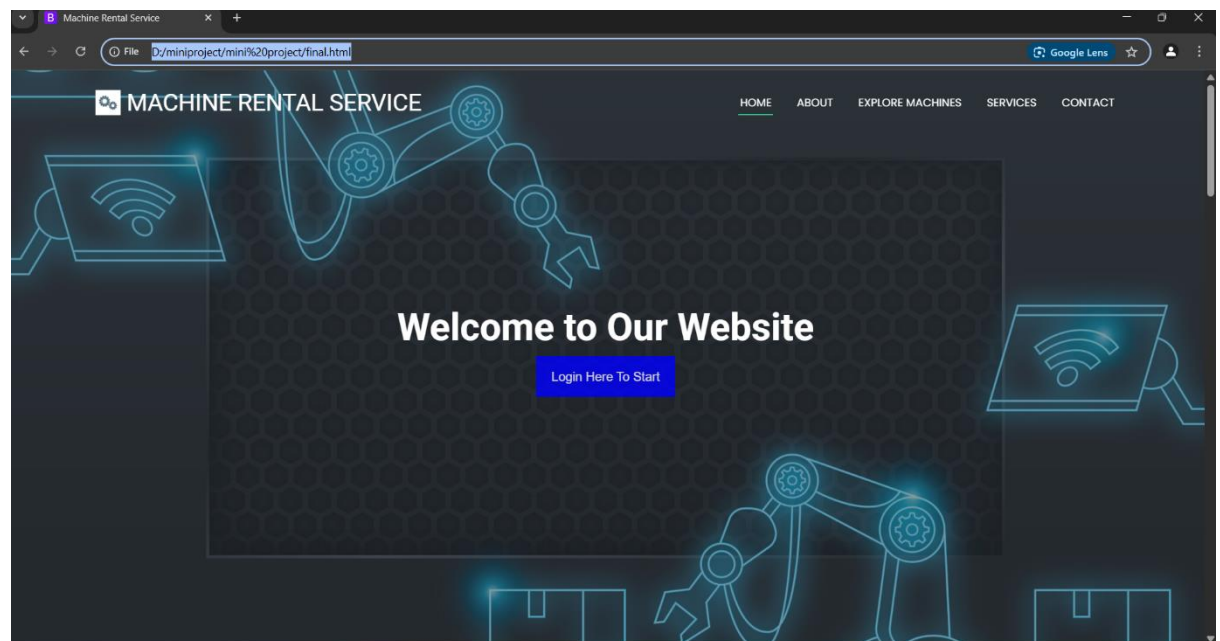
Step 7: Logout or Continue

User can log out.

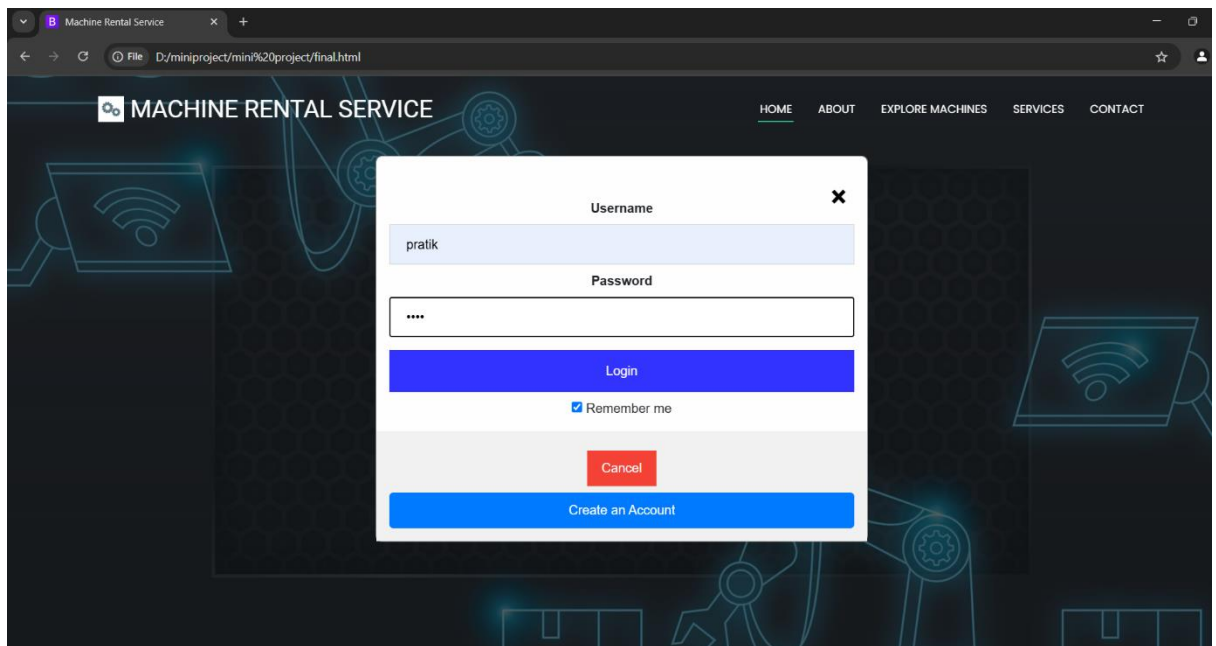
Or continue using the system to explore more machines or add more listings.

User Manual:

1.Home Page:

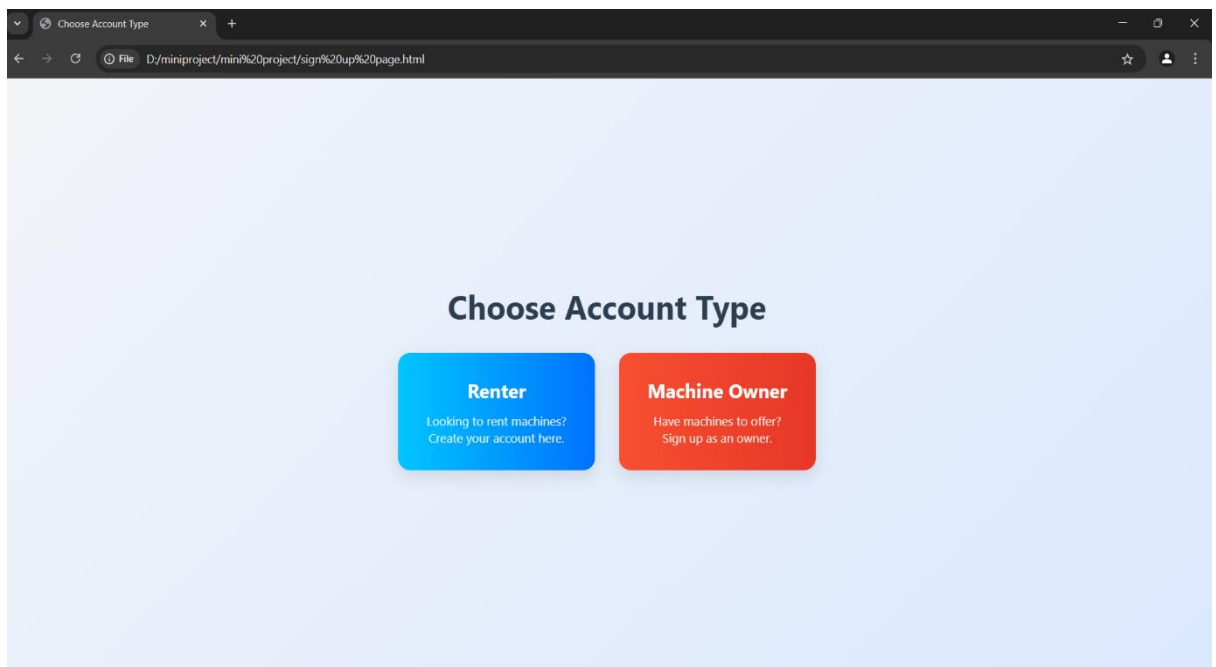


2.Login Form :



The screenshot shows a web browser window with the title "Machine Rental Service". The address bar displays "D:/miniproject/min%20project/final.html". The website has a dark blue header with the logo and navigation links: HOME, ABOUT, EXPLORE MACHINES, SERVICES, and CONTACT. A modal login form is centered on the screen. The form has a title "Username" with a close button (X). It contains two input fields: "Username" with the text "pratik" and "Password" with masked characters "****". Below the password field is a blue "Login" button, a checkbox labeled "Remember me" which is checked, a red "Cancel" button, and a blue "Create an Account" button.

3.Select Your Role :



The screenshot shows a web browser window with the title "Choose Account Type". The address bar displays "D:/miniproject/min%20project/sign%20up%20page.html". The page has a light blue background. In the center, there is a heading "Choose Account Type". Below the heading are two buttons: a blue "Renter" button and a red "Machine Owner" button. The "Renter" button has the text "Looking to rent machines? Create your account here." and the "Machine Owner" button has the text "Have machines to offer? Sign up as an owner."

4. Renter Signup :

Renter Signup

Full Name
payal

Email Address
paya@123gmail.com

Password
123456

Contact Number
8799126898

City
kolhapur

Purpose of Renting (optional)
Personal use

Submit

Account is created Successfully!!

5. Storing renter data into google sheet :

	A	B	C	D	E	F	G	H	I
	name	email	password	contact	city	purpose			
2	pradnya sutar	paya@123gmail.com	1234	8799126898	ichalkaranji	gyfswa4tlg			
3	pradnya sutar	sgdywfty@gmail.com	1234	13879856879	mumbai	new one			
4	payal	paya@123gmail.com	1234	8799126898	ichalkaranji	Personal use			
5	payal	paya@123gmail.com	123456	8799126898	kolhapur	Personal use			
6	payal	paya@123gmail.com	123456	8799126898	kolhapur	Personal use			
7	payal	paya@123gmail.com	123456	8799126898	kolhapur	Personal use			
8	payal	paya@123gmail.com	123456	8799126898	kolhapur	Personal use			
9	payal	paya@123gmail.com	123456	8799126898	kolhapur	Personal use			
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									
24									
25									

6.Owner Signup :

Owner Signup

Full Name
pratik

Email Address
paya@123gmail.com

Password
aghtdw

Contact Number
13879856879

City
ichalkaranji

Add Machine Details

Machine Name
jcb

Machine Type
Construction

Machine Description
new one

Submit

Account is created Successfully!

7.Storing Machine Owner data into google sheet :

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	name	email	password	contact	city	machine-name	machine-type	machine-desc					
2	mrudula waichal	muduj@123gmail.com	xyz	7823569412	solhapur	bulldozer	construction	model:123cdfgh					
3	payal	paya@123gmail.com	123456	8799126898	mumbai	jcb	construction	new one					
4	pratik	paya@123gmail.com	aghtdw	13879856879	ichalkaranji	jcb	construction	new one					
5													
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6.Integration and Testing

Integration and Testing

This phase involves combining all individual modules (login, sign-up, machine upload, explore, booking) and ensuring they work together as a complete system. It also includes testing for functionality, performance, and reliability.

6.1. Integration Process :

Frontend Integration

Linked HTML forms with backend processing (PHP, Python, or Node.js).

Connected pages: Home → Login → Signup → Role-specific dashboards.

Backend Integration

Integrated form submissions with database (MySQL).

Created endpoints to handle login, signup, and machine actions.

Database Integration

Connected application to MySQL database using proper credentials.

Ensured data is stored, retrieved, and updated accurately.

6.2. Testing Types Performed :

Test Type	Description
Unit Testing	Each module (login, signup, add machine, explore) was tested individually.
Integration Testing	Verified how modules interact together (e.g., signup → login → dashboard).
Functional Testing	Ensured all features work correctly (login validation, machine upload, etc).
UI/UX Testing	Checked layout, button behavior, modal displays, and responsiveness.
Security Testing	Prevented unauthorized access, SQL injection, and form bypassing.
Error Handling	Checked for empty fields, invalid login, and database connection errors.

6.3.Sample Test Cases

Test Case	Input	Expected Result	Output	Pass/Fail
Login with valid credentials	Login button	Redirect to Select Role	Login successfully	Pass
Login with invalid credentials	Login Button	Show error message	Throw error message	Pass
Sign up as Owner	Sign up	Redirect to owner form	Account is created	Pass
Sign up as Renter	Sign up	Redirect to Renter form	Account is created	pass
Storing of owner data	Submit Button	Store all entered data	Data is stored into a goggle sheet	Pass
Storing of Renter data	Submit Button	Store all entered data	Data is stored into a goggle sheet	Pass
Access page without login		Redirect to login page		Pass

7.Future Scope

The Machine Rental Service project, though currently focused on being a simple listing and contact-exchange platform, has considerable potential for future enhancements. These can significantly broaden its impact and usability:

1. **Integration of Payment Gateway:** Introducing secure online payments could make the platform a complete rental solution, allowing transactions within the platform.
2. **Smart Matchmaking Algorithms:** Implementing machine learning to suggest optimal matches between renters and machines based on location, usage history, or rental trends.
3. **Mobile Application:** Developing an Android/iOS app version to reach a wider audience and provide on-the-go access.
4. **GPS-based Location Services:** Enabling live machine location tracking for better logistics and trust between parties.
5. **Review and Rating System:** Allowing users to rate and review machines and renters/owners to build credibility and transparency.
6. **Automated Notifications:** Adding email/SMS reminders for rental periods, availability, or interest in listings.
7. **Rental Agreement Templates:** Offering downloadable or e-sign rental agreement templates to formalize deals without legal complexities.
8. **Blockchain for Transparency:** Integrating blockchain to create immutable rental logs and improve trust in transactions.
9. **B2B Expansion:** Tailoring services for industries like mining, farming, or construction, potentially including fleet management.
10. **Data Analytics Dashboard:** Providing analytics on usage trends, machine demand, and rental performance for strategic insights.
11. **Smart Matchmaking Algorithms:** Implementing machine learning to suggest optimal matches between renters and machines based on location, usage history, or rental trends.

8.Applications

The Machine Rental Service platform has broad applicability across numerous sectors where machines are essential but not always required permanently:

Construction Industry: For renting heavy equipment like bulldozers, cranes, and concrete mixers.

Agriculture Sector: Farmers can rent tractors, harvesters, or irrigation equipment during specific seasons.

Small-Scale Industries: Businesses can lease CNC machines, packaging systems, or printing equipment as needed.

Event Management: Renting out sound systems, lighting rigs, or generators for temporary use.

Disaster Response & Relief: Rapid deployment of essential machines to areas affected by natural disasters.

Municipal Utilities: Local governments can rent cleaning and maintenance machinery for short-term civic operations.

Remote Locations: The platform can serve users in rural or remote areas where machine procurement is difficult.

Academic & Research Institutions: Laboratories can rent out scientific machines for field research and temporary projects.

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